

Annex B (normative)

Interpretation and notation of failure and maintenance parameters

B.1 Failure interpretation

When planning to collect data (see 7.1.2 and B.2.6), be aware that a failure can occur in one of a number of failure modes, e.g. complete loss of function, function degradation below an acceptable limit or an imperfection in the state or condition of an item (incipient failure) that is likely to result in a functional failure if not corrected.

Also be aware that it can be useful to make a distinction between the collection of data for reliability purposes and for availability purposes as follows.

- a) For reliability purposes, it is mainly the intrinsic failures of the equipment unit that are of interest, i.e. physical failures that occur on the equipment being considered and that normally require some restoration (corrective maintenance) that it is necessary to record.
- b) For the full lifetime story of equipment, it is necessary to record all actual preventive maintenance actions in a way similar to that for the corrective maintenance.
- c) For availability purposes, all failures that have caused some equipment outage should be recorded. This can include stoppages due to operational limits being exceeded (e.g. trips) where no physical failure of the equipment occurred.
- d) Even if no failures are experienced within the surveillance time, it is possible to estimate the failure rate by properly censored data (see C.3.3). Hence, recording the reliability history may also be useful for equipment in periods with no failures.

Table B.1 gives some guidance on this issue by distinguishing between data collected as reliability data and additional data collected as availability data.

Annex F and IEC 61508 also give guidance on what to consider as a failure for safety equipment. Such definition can be related to functional loss, reduced capacity or operation outside prescribed limits.

The full description of a failure might not be possible before a corrective action is carried out. In some cases (incipient failures), the corrective action may deliberately be deferred (e.g. opportunity maintenance). In this case, it can be necessary to record both the date of failure detection and the date of the corrective action. For analysis purposes, the latter date should normally be used.

Table B.1 — Failure in relation to reliability and availability

Type of failure/maintenance to record	Reliability	Availability
Failures that require some corrective maintenance action to be carried out (repair, replacement)	Yes	Yes
Failure discovered during inspection, testing and/or preventive maintenance that requires repair or replacement of typically non-wear items (seals, bearings, impellers, etc.)	Yes	Yes
Failure of safety devices or control/monitoring devices that necessitates shutdown (trip) or reduction of the items capability below specified limits	Yes	Yes
Shutdown (trip) of the item (whether automatically or manually controlled) due to external conditions or operating errors, where no physical failure condition of the item is revealed	No	Yes
Failure of the equipment caused by external impact (e.g. lack of power supply, structural impact, etc.)	No	Yes
Periodic replacement of consumables and normal wear parts	No	No
Minor planned maintenance services, such as adjustments, lubrication, cleaning, oil replacement, filter replacement or cleaning, painting, etc.	No	Yes
Testing and inspections	No	Yes
"On-demand" activations	Yes	Yes
Preventive or planned maintenance ^a	Yes (No)	Yes
Modifications, new work, upgrades ^b	No	Yes/No
^a To get the full lifetime history of the equipment, the actual preventive maintenance should be recorded. For recording failures only, this can be skipped.		
^b Modifications are normally not a part of maintenance but are frequently done by maintenance personnel.		

B.2 Failure and maintenance data notations

B.2.1 General

In order to limit database size and make it easier to analyse the data, it is recommended that coded information be used wherever applicable. A drawback with codes is that potentially useful information can be lost and that selecting inappropriate codes can lead to non-informative information. The availability of too many codes can be confusing and the codes can overlap, while too few codes might not sufficiently describe the area one is aiming to cover. A unified definition and interpretation of codes is necessary for obtaining highly reliable information.

In all cases, it is recommended to supplement the coding with some additional free-text capability in order to improve interpretation of single events, both for quality purposes before the data are entered into the database and for subsequent detailed analysis of single records (e.g. failure events).

Annex B.2 presents a method of coding that has been found to be useful when collecting RM data within the petroleum and natural gas industry, and should be equally applicable for similar equipment classes in the petrochemical industry. For some specific equipment and/or specific uses, supplementary codes may be used.

Design a method of reporting failure (see 7.1.2) that records the time and date of failure together with details of the failure mode (see B.2.6), the failure mechanism (see B.2.2) and the failure cause (root cause) (see B.2.3). Also, record the detection method (see B.2.4) and the maintenance activity (see B.2.5). Use the codes given in the tables wherever practicable and additional free text where necessary.

Take care to distinguish between failure mechanism and failure mode.

Failure modes are presented in this Annex B in Tables B.6 to B.12 for those equipment examples included in Annex A as shown in Table A.4.

Subdivision codes for failure mechanisms and failure causes, e.g. numbers 1.1, 1.2, etc., should be preferred before the general category failure code, e.g. 1, and so on (see Tables B.2. and B.3).

How failure mode, failure mechanism and failure cause are related to different taxonomy levels is shown in Table 3.

B.2.2 Failure mechanism

The failure mechanism is the physical, chemical or other process or combination of processes that leads to the failure. It is an attribute of the failure event that can be deduced technically, e.g. the apparent, observed cause of the failure. The failure mechanism's root cause(s) is/are coded whenever this information is available. (A separate field for this is recommended in this International Standard.)

The codes on failure mechanism are basically related to one of the following major categories of failure types:

- a) mechanical failures;
- b) material failures;
- c) instrumentation failures;
- d) electrical failures;
- e) external influence;
- f) miscellaneous.

This categorization is rather coarse and within each category a more detailed categorization is recommended as shown in Table B.2. If there is not sufficient information to apply codes at this sublevel, then codes on the main level as listed above may be used. This implies that descriptive codes for mechanical failures, numbered 1.1, 1.2, etc., should be preferred to the general category failure code, 1.0, and so on (see Table B.2).

The failure mechanism should normally be related to a lower indenture level (subunit or maintainable-item level). In practical terms, the failure mechanism represents a failure mode at maintainable item level.

Care should be taken to distinguish between failure mechanism and failure mode.

EXAMPLE It is recorded that a valve started leaking hydrocarbons to the environment but no further causes are recorded. Here, the failure mode should be coded ELP (external leak of process medium) and the failure mechanism coded unknown (6.4), not leakage (1.1).

Failure mechanism is also related to the failure cause (see B.2.3); the latter aimed at revealing the underlying root cause of the failure.

Six categories of failure mechanism are identified in Table B.2, together with subdivisions and related codes to be used in data bases.

Table B.2 — Failure mechanism

Failure mechanism		Subdivision of the failure mechanism		Description of the failure mechanism
Code number	Notation	Code number	Notation	
1	Mechanical failure	1.0	General	A failure related to some mechanical defect but where no further details are known
		1.1	Leakage	External and internal leakage, either liquids or gases: If the failure mode at equipment unit level is coded as "leakage", a more causally oriented failure mechanism should be used wherever possible.
		1.2	Vibration	Abnormal vibration: If the failure mode at equipment level is vibration, which is a more causally oriented failure mechanism, the failure cause (root cause) should be recorded wherever possible.
		1.3	Clearance/alignment failure	Failure caused by faulty clearance or alignment
		1.4	Deformation	Distortion, bending, buckling, denting, yielding, shrinking, blistering, creeping, etc.
		1.5	Looseness	Disconnection, loose items
		1.6	Sticking	Sticking, seizure, jamming due to reasons other than deformation or clearance/alignment failures
2	Material failure	2.0	General	A failure related to a material defect but no further details known
		2.1	Cavitation	Relevant for equipment such as pumps and valves
		2.2	Corrosion	All types of corrosion, both wet (electrochemical) and dry (chemical)
		2.3	Erosion	Erosive wear
		2.4	Wear	Abrasive and adhesive wear, e.g. scoring, galling, scuffing, fretting
		2.5	Breakage	Fracture, breach, crack
		2.6	Fatigue	If the cause of breakage can be traced to fatigue, this code should be used.
		2.7	Overheating	Material damage due to overheating/burning
3	Instrument failure	2.8	Burst	Item burst, blown, exploded, imploded, etc.
		3.0	General	Failure related to instrumentation but no details known
		3.1	Control failure	No, or faulty, regulation
		3.2	No signal/indication/alarm	No signal/indication/alarm when expected
		3.3	Faulty signal/indication/alarm	Signal/indication/alarm is wrong in relation to actual process. Can be spurious, intermittent, oscillating, arbitrary
		3.4	Out of adjustment	Calibration error, parameter drift
		3.5	Software failure	Faulty, or no, control/monitoring/operation due to software failure
		3.6	Common cause/mode failure	Several instrument items failed simultaneously, e.g. redundant fire and gas detectors; also failures related to a common cause.

Table B.2 (continued)

Failure mechanism		Subdivision of the failure mechanism		Description of the failure mechanism
Code number	Notation	Code number	Notation	
4	Electrical failure	4.0	General	Failures related to the supply and transmission of electrical power, but where no further details are known
		4.1	Short circuiting	Short circuit
		4.2	Open circuit	Disconnection, interruption, broken wire/cable
		4.3	No power/voltage	Missing or insufficient electrical power supply
		4.4	Faulty power/voltage	Faulty electrical power supply, e.g. overvoltage
		4.5	Earth/isolation fault	Earth fault, low electrical resistance
5	External influence	5.0	General	Failure caused by some external events or substances outside the boundary but no further details are known
		5.1	Blockage/plugged	Flow restricted/blocked due to fouling, contamination, icing, flow assurance (hydrates), etc.
		5.2	Contamination	Contaminated fluid/gas/surface, e.g. lubrication oil contaminated, gas-detector head contaminated
		5.3	Miscellaneous external influences	Foreign objects, impacts, environmental influence from neighbouring systems
6	Miscellaneous ^a	6.0	General	Failure mechanism that does not fall into one of the categories listed above
		6.1	No cause found	Failure investigated but cause not revealed or too uncertain
		6.2	Combined causes	Several causes: If there is one predominant cause this should be coded.
		6.3	Other	No code applicable: Use free text.
		6.4	Unknown	No information available

^a The data acquirer should judge which is the most important failure mechanism descriptor if more than one exist, and try to avoid the 6.3 and 6.4 codes.

B.2.3 Failure cause

The objective of these data is to identify the initiating event ("root causes") in the sequence leading up to a failure of an equipment item. Five categories of failure cause are identified in Table B.3 together with sub divisions and related codes to be used in data bases.

The failure causes are classified in the following categories:

- 1) design-related causes;
- 2) fabrication/installation-related causes;
- 3) failures related to operation/maintenance;
- 4) failures related to management;
- 5) miscellaneous.

As for failure mechanism, the failure cause can be recorded at two levels depending on how much information is available. If the information is scarce, only a coarse classification, i.e. codes 1, 2, 3, 4 and 5, can be possible, while a more detailed subdivision code number can be recorded if more information is available.

Failure causes are commonly not known in depth when the failure is observed and, in order to reveal the root cause of a failure, a specific root cause analysis can be useful. This is in particular relevant for failures of a more complex nature and where the failure is important to avoid due to its consequences. Examples are failures with serious safety and/or environmental consequences, abnormally high failure rates compared to the average and failures with a high repair cost.

Due care is required so as not to confuse failure mechanism (describing the apparent, observed cause of failure) with failure cause (describing the underlying or "root" cause of a failure).

Table B.3 — Failure causes

Code number	Notation	Subdivision code number	Subdivision of the failure cause	Description of the failure cause
1	Design-related causes	1.0	General	Inadequate equipment design or configuration (shape, size, technology, configuration, operability, maintainability, etc.), but no further details known
		1.1	Improper capacity	Inadequate dimensioning/capacity
		1.2	Improper material	Improper material selection
2	Fabrication/ installation-related causes	2.0	General	Failure related to fabrication or installation, but no further details known
		2.1	Fabrication error	Manufacturing or processing failure
		2.2	Installation error	Installation or assembly failure (assembly after maintenance not included)
3	Failure related to operation/ maintenance	3.0	General	Failure related to operation/use or maintenance of the equipment but no further details known
		3.1	Off-design service	Off-design or unintended service conditions, e.g. compressor operation outside envelope, pressure above specification, etc.
		3.2	Operating error	Mistake, misuse, negligence, oversights, etc. during operation
		3.3	Maintenance error	Mistake, errors, negligence, oversights, etc. during maintenance
		3.4	Expected wear and tear	Failure caused by wear and tear resulting from normal operation of the equipment unit
4	Failure related to management	4.0	General	Failure related to management issues, but no further details known
		4.1	Documentation error	Failure related to procedures, specifications, drawings, reporting, etc.
		4.2	Management error	Failure related to planning, organization, quality assurance, etc.
5	Miscellaneous ^a	5.0	Miscellaneous - general	Causes that do not fall into one of the categories listed above
		5.1	No cause found	Failure investigated but no specific cause found
		5.2	Common cause	Common cause/mode
		5.3	Combined causes	Several causes are acting simultaneously. If one cause is predominant, this cause should be highlighted.
		5.4	Other	None of the above codes applies. Specify cause as free text.
		5.5	Unknown	No information available related to the failure cause

^a The data acquirer should judge which is the most important cause if more than one exist, and try to avoid the 5.4 and 5.5 codes.

B.2.4 Detection method

This is the method or activity by which a failure is discovered. This information is vitally important when evaluating the effect of maintenance, e.g. to distinguish between failures discovered by a planned action (inspection, PM maintenance) or by chance (casual observation). Nine categories of detection methods are identified in Table B.4, together with related codes to be used in the databases.

Table B.4 — Detection method

Number	Notation ^a	Description	Activity
1	Periodic maintenance	Failure discovered during preventive service, replacement or overhaul of an item when executing the preventive maintenance programme	Scheduled activities
2	Functional testing	Failure discovered by activating an intended function and comparing the response against a predefined standard. This is one typical method for detecting hidden failures	
3	Inspection	Failure discovered during planned inspection, e.g. visual inspection, non-destructive testing	
4	Periodic condition monitoring ^b	Failures revealed during a planned, scheduled condition monitoring of a predefined failure mode, either manually or automatically, e.g. thermography, vibration measuring, oil analysis, sampling	
5	Continuous condition monitoring ^b	Failures revealed during a continuous condition monitoring of a predefined failure mode	Continuous monitoring
6	Production interference	Failure discovered by production upset, reduction, etc.	
7	Casual observation	Casual observation during routine or casual operator checks, mainly by senses (noise, smell, smoke, leakage, appearance etc.)	Casual occurrences
8	Corrective maintenance	Failure observed during corrective maintenance	
9	On demand	Failure discovered during an on-demand attempt to activate an equipment unit (e.g. safety valve fails to close on ESD-signal, fail to start a gas turbine on demand, etc.)	
10	Other	Other observation method and/or combination of several methods	Other

^a Specific notation for fire and gas detectors, process sensors and control logic units: The codes above should be interpreted as follows:

functional test	periodic functional testing
casual observation	field observation
periodic CM	abnormal state discovered by control room personnel (no fault annunciation)
continuous CM	fault annunciation in control room (audible and/or visible alarm)

^b Condition monitoring implies use of specific equipment and/or algorithms to monitor the condition of the equipment with respect to predefined failure modes (note that "test" and "inspection" are separate codes). Condition monitoring (CM) can be further divided into either 1) periodic monitoring or 2) continuous monitoring as follows:

- 1) periodic CM: periodic condition monitoring includes techniques such as thermography, off-line vibration measuring, oil analyses, calibration checks and sampling;
- 2) continuous CM: continuous instrumental surveillance of process parameters and equipment condition, e.g. temperature, pressure, flow, RPM, to detect abnormal operating conditions.

B.2.5 Maintenance activity

Twelve categories of maintenance activity are identified in Table B.5 together with related codes to be used in databases for both corrective maintenance and preventive maintenance.

Table B.5 — Maintenance activity

Code Number	Activity	Description	Examples	Use ^a
1	Replace	Replacement of the item by a new or refurbished item of the same type and make	Replacement of a worn-out bearing	C, P
2	Repair	Manual maintenance action performed to restore an item to its original appearance or state	Repack, weld, plug, reconnect, remake, etc.	C
3	Modify ^b	Replace, renew or change the item, or a part of it, with an item/part of a different type, make, material or design	Install a filter with smaller mesh diameter, replace a lubrication oil pump with another type, reconfiguration etc.	C, P
4	Adjust	Bringing any out-of-tolerance condition into tolerance	Align, set and reset, calibrate, balance	C, P
5	Refit	Minor repair/servicing activity to bring back an item to an acceptable appearance, internal and external	Polish, clean, grind, paint, coat, lube, oil change, etc.	C, P
6	Check ^c	The cause of the failure is investigated, but no maintenance action performed, or action is deferred. Able to regain function by simple actions, e.g. restart or resetting.	Restart, resetting, no maintenance action, etc. Particularly relevant for functional failures, e.g. fire and gas detectors, subsea equipment	C
7	Service	Periodic service tasks: Normally no dismantling of the item	e.g. cleaning, replenishment of consumables, adjustments and calibrations	P
8	Test	Periodic test of function or performance	Function test of gas detector, accuracy test of flow meter	P
9	Inspection	Periodic inspection/check: a careful scrutiny of an item carried out with or without dismantling, normally by use of senses	All types of general check. Includes minor servicing as part of the inspection task	P
10	Overhaul	Major overhaul	Comprehensive inspection/overhaul with extensive disassembly and replacement of items as specified or required	C, P
11	Combination	Several of the above activities are included	If one activity dominates, this may alternatively be recorded	C, P
12	Other	Maintenance activity other than specified above	may dominates	C, P
^a C: used typically in corrective maintenance; P: used typically in preventive maintenance. ^b Modification is not defined as a maintenance category, but is often performed by persons trained in the maintenance disciplines. Modification to a major extent can have influence on the operation and reliability of an equipment unit. ^c "Check" includes the circumstances both where a failure cause was revealed but maintenance action was considered either not necessary or not possible to carry out and where no failure cause could be found.				

For corrective maintenance, this information describes the type of restoration action that was performed. In general, the predominant restoration activity should be coded when several activities are involved. The code categories "repair", "replace", "overhaul" and "modify" should have a priority relative to the code categories "refit" and "adjust" when a combination of the two categories are involved (e.g. repair consisting of "repair" and "refit" should be coded as "repair"). If there are several repair activities involved, none of which is predominant, the code "combined" may be used.

"Modify" means a modification of the original equipment unit where the original design has been altered or the item in question replaced with one of a different type/make. If the modification is of significant character, it is not considered as a maintenance action, but may be carried out by, or in co-operation with, the maintenance staff. A "repair" is meant to be an action to correct a single failure or a few failures, normally on-site. "Overhaul" means a comprehensive repair of several failures, or one major failure requiring extensive work, or complete refurbishment of an equipment subunit. Typically, such maintenance is undertaken in a workshop.

If the complete equipment unit has been replaced with a new and/or modified one, it is recommended to rewind the time parameters (e.g. operating time) for this unit. This does not apply if the equipment unit is of low complexity and a complete replacement is considered as a normal part of the maintenance.

For preventive maintenance, this information describes the type of preventive action being performed. In general, the most predominant maintenance activity should be coded when several activities are involved. If there is no predominant task, again this should be coded as "combined" and additional information on the various activities listed in a free-text field if provided.

NOTE These maintenance codes do not, as such, reflect the effectiveness of the maintenance action as to restoring the condition of the item (e.g. "good-as-new" or "bad-as-old" condition).

B.2.6 Failure modes

Failure modes should normally relate to the equipment-class level in the hierarchy. For subsea equipment, however, it is recommended to also record failure modes on lower levels in the equipment hierarchy (e.g. "maintainable-item" level). The failure modes can be categorized into three types:

- a) desired function is not obtained (e.g. failure to start);
- b) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- c) failure indication is observed but there is no immediate and critical impact on the equipment-unit function [these are typically non-critical failures related to some degradation or incipient fault condition (e.g. initial wear)].

Failure modes are presented in Tables B.6 to B.12 for each main equipment category shown in Table A.4.

Recommended failure modes are presented for each main equipment category (see also list of equipment presented in Table A.4):

- rotating (compressors, combustion engines, electric generators, gas turbines, etc.);
- mechanical (cranes, heat exchangers, heaters and boilers, vessels, storage tanks, piping, etc.);
- electrical (UPS, power transformers, frequency converters, etc.);
- safety and control (fire and gas detectors, sensors, valves, nozzles, fire fighting equipment, etc.);
- subsea production (subsea control systems, Xmas trees, templates, manifolds, risers, etc.);
- well completion (downhole safety valves, wellheads, tubing, casing, packers, etc.);
- drilling (derrick, top drive, drawworks, mud pump, BOP, etc.).

Table B.6 — Rotating equipment — Failure modes

Equipment class ^a								Failure modes			
Combustion engine	Compressor	Electric generator	Electric motor	Gas turbine	Pump	Steam turbine	Turbo expander	Description	Examples	Code ^b	Type ^c
X	X	X	X	X	X	X	X	Failure to start on demand	Doesn't start on demand	FTS	1
X	X	X	X					Failure to stop on demand	Doesn't stop on demand	STP	1
X	X	X	X	X	X	X	X	Spurious stop	Unexpected shutdown	UST	2
X	X	X	X	X	X	X	X	Breakdown	Serious damage (seizure, breakage)	BRD	3
X	X		X	X	X	X	X	High output	Overspeed/output above acceptance	HIO	2
X	X	X	X	X	X	X	X	Low output	Delivery/output below acceptance	LOO	2
X	X		X	X	X	X	X	Erratic output	Oscillating, hunting, instability	ERO	2
X				X		X		External leakage – fuel	External leakage of supplied fuel/gas	ELF	3
	X			X	X	X	X	External leakage – process medium	Oil, gas, condensate, water	ELP	3
X	X	X	X	X	X	X	X	External leakage – utility medium	Lubricant, cooling water	ELU	3
X	X			X	X	X	X	Internal leakage	Leakage internally of process or utility fluids	INL	3
X	X	X	X	X	X	X	X	Vibration	Abnormal vibration	VIB	3
X	X	X	X	X	X	X	X	Noise	Abnormal noise	NOI	3
X	X	X	X	X	X	X	X	Overheating	Machine parts, exhaust, cooling water	OHE	3
X	X			X	X	X	X	Plugged/choked	Flow restriction(s)	PLU	3 (2)
X	X	X	X	X	X	X	X	Parameter deviation	Monitored parameter exceeding limits, e.g. high/low alarm	PDE	2 (3)

Table B.6 (continued)

Equipment class ^a								Failure modes			
Combustion engine	Compressor	Electric generator	Electric motor	Gas turbine	Pump	Steam turbine	Turbo expander	Description	Examples	Code ^b	Type ^c
X	X	X	X	X	X	X	X	Abnormal instrument reading	False alarm, faulty instrument indication	AIR	2 (3)
X	X	X	X	X	X	X	X	Structural deficiency	Material damages (cracks, wear, fracture, corrosion)	STD	3
X	X	X	X	X	X	X	X	Minor in-service problems	Loose items, discoloration, dirt	SER	3
X	X	X	X	X	X	X	X	Other	Failure modes not covered above	OTH	—
X	X	X	X	X	X	X	X	Unknown	Too little information to define a failure mode	UNK	—
^a See Table A.4. The codes shown apply to equipment classes marked with "X". ^b A proposed abbreviated code for the failure-mode. ^c One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment): 1) desired function is not obtained (e.g. failure to start); 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output); 3) failure indication is observed, but there is no immediate and critical impact on equipment-unit function. These are typically non-critical failures related to some degradation or incipient fault condition.											

Table B.7 — Mechanical Equipment — Failure modes

Equipment class ^a								Failure modes			
Cranes	Heat exchangers	Heaters and boilers	Piping	Vessels	Winches	Turrets	Swivels	Description	Examples	Code ^b	Type ^c
X	X	X	X	X	X	X	X	Abnormal instrument reading	False alarm, faulty instrument indication	AIR	2 (3)
X			X		X			Breakdown	Breakdown	BRD	3 (1)
	X			X				Insufficient heat transfer	Cooling/heating below acceptance	IHT	2
	X	X	X	X			X	External leakage – process medium	Oil, gas, condensate, water	ELP	3
X	X	X		X	X		X	External leakage – utility medium	Lubricant, cooling water, barrier oil	ELU	3
						X	X	Failure to connect	Failure to connect	FCO	1
X						X	X	Failure to function as intended	General operation failure	FTI	1(2)
X					X	X	X	Failure to rotate	Failure to rotate	FRO	1
X					X			Failure to start on demand	Failure to start on demand	FTS	1
					X			Failure to stop on demand	Failure to stop on demand	STP	1
						X		Failure to disconnect	Failure to disconnect when demanded	FDC	2
		X						Insufficient heat transfer	Missing, or too low, heat transfer	IHT	2
X	X	X	X				X	Internal leakage	Leakage internally of process or utility fluids	INL	3
							X	Low oil supply pressure	Low oil supply pressure	LBP	2
					X			Low output	Performance below specifications	LOO	2
X					X			Load drop	Load drop	LOA	2
						X		Loss of buoyancy	Loss of buoyancy in idle position	LOB	2
						X		Mooring failure	Mooring failure	MOF	2

Table B.7 (continued)

Equipment class ^a								Failure modes			
Cranes	Heat exchangers	Heaters and boilers	Piping	Vessels	Winches	Turrets	Swivels	Description	Examples	Code ^b	Type ^c
X			X		X	X		Noise	Excessive noise	NOI	3
X		X	X		X			Overheating	Overheating	OHE	3
	X	X	X	X			X	Plugged/choked	Flow restriction due to contamination, objects, wax, etc.	PLU	3
			X				X	Power/signal transmission failure	Power/signal transmission failure	PTF	2
X					X			Slippage	Wire slippage	SLP	2
X					X			Spurious operation	Unexpected operation	SPO	2
X	X	X	X	X	X	X	X	Structural deficiency	Material damages (cracks, wear, fracture, corrosion)	STD	3
X	X	X	X	X	X	X	X	Parameter deviation	Monitored parameter exceeding limits, e.g. high/low alarm	PDE	2 (3)
X			X		X			Vibration	Excessive vibration	VIB	3
X	X	X	X	X	X	X	X	Minor in-service problems	Loose items, discoloration, dirt	SER	3
X	X	X	X	X	X	X	X	Other	Failure modes not covered above	OTH	—
X	X	X	X	X	X	X	X	Unknown	Too little information to define a failure mode	UNK	—

^a See Table A.4. The codes shown apply to equipment classes marked with "X".

^b A proposed abbreviated code for the failure mode.

^c One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment):

- 1) desired function is not obtained (e.g. failure to start);
- 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- 3) failure indication is observed, but there is no immediate and critical impact on equipment-unit function. These are typically non-critical failures related to some degradation or incipient fault condition.

Table B.8 — Electrical equipment — Failure modes

Equipment class ^a		Failure modes			
UPS	Power transformers	Description	Examples	Code ^b	Type ^c
X	X	Failure to function on demand	Doesn't start on demand	FTF	1
X		Faulty output frequency	Wrong/oscillating frequency	FOF	2
X	X	Faulty output voltage	Wrong/unstable output voltage	FOV	2
X		Loss of redundancy	One or more redundant units not functioning	LOR	2
X		Erratic output	Oscillating, hunting, instability	ERO	2
X	X	Overheating	Machine parts, exhaust, cooling water	OHE	3
X	X	Parameter deviation	Monitored parameter exceeding limits, e.g. high/low alarm	PDE	2 (3)
X		Spurious operation	Unexpected operation	SPO	2
	X	Abnormal instrument reading	Wrong oil level indication	AIR	3
	X	Plugged/choked	Obstructed piping	PLU	2
	X	External leakage utilities	Oil leakage	ELU	1
	X	Structural deficiency	Reservoir rupture	STD	1
	X	Internal leakage	Oil leakage	INL	2
X	X	Minor in-service problems	Loose items, discoloration, dirt	SER	3
X	X	Other	Failure modes not covered above	OTH	—
X	X	Unknown	Too little information to define a failure mode	UNK	—

^a See Table A.4. The codes shown apply to equipment classes marked with "X".

^b A proposed abbreviated code for the failure mode.

^c One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment):

- 1) desired function is not obtained (e.g. failure to start);
- 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- 3) failure indication is observed, but there is no immediate and critical impact on equipment-unit function. These are typically non-critical failures related to some degradation or incipient fault condition.

Table B.9 — Safety and control equipment — Failure modes

Equipment class ^a					Failure modes			
Fire detectors ^b	Gas detectors ^b	Input devices	Control logic units	Valves	Description	Examples	Code ^c	Type ^d
X		X	X		Failure to function on demand	Failure to respond on signal/activation	FTF	1
				X	Failure to open on demand	Doesn't open on demand	FTO	1
				X	Failure to close on demand	Doesn't close on demand	FTC	1
				X	Delayed operation	Opening/closing time below spec.	DOP	2
X	X	X	X	X	Spurious operation	e.g. false alarm	SPO	2
X	X ^e	X	X	X	High output	Overspeed/output above acceptance	HIO	2
X	X ^f	X	X	X	Low output	Delivery/output below acceptance	LOO	2
	X ^g				Very low output		VLO	2
X		X	X		Erratic output	Oscillating, hunting, instability	ERO	2
X	X ^h	X			No output	No output	NOO	1
X	X				Spurious high alarm level	e.g. 60 % of Lower Explosion Limit (LEL)	SHH	2
X	X				Spurious low alarm level	e.g. 20 % of Lower Explosion Limit (LEL)	SLL	2
				X	Plugged/choked	Partial or full flow restriction	PLU	1
		X		X	External leakage – process medium	Oil, gas, condensate, water	ELP	3
		X		X	External leakage – utility medium	Lubricant, cooling water	ELU	3
				X	Internal leakage	Leakage internally of process or utility fluids	INL	3
				X	Leakage in closed position	Leak through valve in closed position	LCP	
				X	Abnormal instrument reading	False alarm, faulty instrument indication	AIR	2 (3)

Table B.9 (continued)

Equipment class ^a					Failure modes			
Fire detectors ^b	Gas detectors ^b	Input devices	Control logic units	Valves	Description	Examples	Code ^c	Type ^d
				X	Structural deficiency	Material damages (cracks, wear, fracture, corrosion)	STD	3
X		X	X	X	Minor in-service problems	Loose items, discoloration, dirt	SER	3
X	X	X		X	Other	Failure modes not covered above	OTH	—
X	X	X	X	X	Unknown	Too little information to define a failure mode	UNK	—

^a See Table A.4. The codes shown apply to equipment classes marked with an "X".

^b Failure coding for fire and gas detectors: For fire and gas detectors, it is important that all failures are recorded; also those detected during scheduled testing and those detected in operation, e.g. replacement of a detector head should be recorded, even if this is done as part of the preventive maintenance programme. Typical failure modes are the following:

- failure to function: The detector does not respond when exposed to its relevant stimulus (e.g. gas or heat). This failure mode is normally observed during functional testing;
- spurious operation: The detector gives an alarm signal when it is not exposed to relevant stimulus. This failure mode is normally observed during operation and logged by control-room personnel;
- others: Additionally, some failure modes related to low/high output, adjustments and overhauls will typically be found in the log books.

^c A proposed abbreviated code for the failure mode.

^d One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment):

- 1) desired function is not obtained (e.g. failure to start);
- 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- 3) failure indication is observed, but there is no immediate and critical impact on equipment-unit function. These are typically non-critical failures related to some degradation or incipient fault condition.

^e e.g. reading 10 % LEL to 20 % LEL without test gas; reading above 80 % LEL on test gas.

^f e.g. reading between 31 % LEL to 50 % LEL upon test gas (assuming a nominal set point of 65 % LEL).

^g e.g. reading between 11 % LEL to 30 % LEL upon test gas.

^h e.g. reading less than 10 % LEL upon test gas.

Table B.10 — Subsea equipment — Failure modes

Equipment class ^b				Failure modes ^a			
Subsea control systems	Xmas trees	Subsea pumps	Risers	Description	Examples	Code ^c	Type ^d
X		X		Failure to function on demand	Failure to respond on signal/activation	FTF	1
	X			Failure to open on demand	Doesn't open on demand	FTO	1
	X			Failure to close on demand	Doesn't close on demand	FTC	1
	X			Failure to lock/unlock	Doesn't lock or unlock when demanded	FTL	1
	X			Failure to set/retrieve	Failed set/retrieve operations	SET	1
X	X	X		Spurious operation	Fails to operate as demanded	SPO	2
		X		High output	Overspeed/output above acceptance	HIO	2
X		X		Low output	Delivery/output below acceptance	LOO	2
X	X			Insufficient power	Lack of or too low power supply	POW	1
X				Loss of redundancy	One or more redundant units failed	LOR	2
	X			Loss of barrier	One or more barriers against oil/gas escape lost	LOB	2
	X		X	Plugged/choked	Partial or full flow restriction	PLU	1
X	X	X	X	External leakage – process medium	Oil, gas, condensate, water	ELP	3
X	X		X	External leakage – utility medium	Lubricant, cooling water	ELU	3
X	X	X	X	Internal leakage – utility medium	Leakage internally of process or utility fluids	INL	3
X		X		Abnormal instrument reading	False alarm, faulty instrument indication	AIR	2 (3)
	X		X	Structural deficiency	Material damages (cracks, wear, fracture, corrosion)	STD	3
X			X	No immediate effect	No effect on function	NON	1
X	X	X	X	Other	Failure modes not covered above	OTH	—

^a Although not a requirement of this International Standard, it is recommended that, for subsea equipment, failure modes are also recorded at a lower hierarchical level, e.g. "maintainable item".

^b See Table A.4. The codes shown apply to equipment classes marked with "X".

^c A proposed abbreviated code for the failure mode.

^d One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment):

- 1) desired function is not obtained (e.g. failure to start);
- 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- 3) failure indication is observed, but there is no immediate and critical impact on equipment-unit function. These are typically non-critical failures related to some degradation or incipient fault condition.

Table B.11 — Well-completion equipment — Failure modes

Equipment class ^a	Failure modes			
	Description	Examples	Code ^b	Type ^c
X	Failure to open on demand	Does not open on demand	FTO	1
X	Failure to close on demand	Does not close upon demand signal	FTC	2
X	Leakage in closed position	Leakage through valve exceeding acceptance criteria when closed	LCP	2
X	Well-to-control-line communication	Influx of well fluids into valve control line	WCL	2
X	Control-line-to-well communication	Loss of hydraulic control fluids into the well bore	CLW	3
X	Premature closure	Spurious closure of valve without command	PCL	2
X	Other	Failure modes not covered above	OTH	—
X	Unknown	Too little information to define a failure mode	UNK	—

^a See Table A.4. The codes shown apply to equipment classes marked with an "X".

^b A proposed abbreviated code for the failure mode.

^c One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment):

- 1) desired function is not obtained (e.g. failure to start);
- 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- 3) failure indication is observed, but there is no immediate and critical impact on equipment unit function. These are typically non-critical failures related to some degradation or incipient fault condition.

Table B.12 — Drilling equipment — Failure modes

Equipment class ^a		Failure modes			
Top drive	Blowout preventer	Description	Examples	Code ^b	Type ^c
	X	Failure to function on demand	Failure to respond on signal/activation (e.g. failure to shear)	FTF	1
	X	Failure to open	Doesn't open on demand	FTO	1
	X	Failure to close	Doesn't close on demand	FTC	1
X	X	Abnormal instrument reading	False alarm, faulty instrument indication	AIR	2 (3)
X	X	External leakage – utility medium	Hydraulic oil, lubrication oil, coolant, mud, water, etc.	ELU	3
X	X	Erratic output	Oscillating or instable operation	ERO	2
X		Failure to start on demand	Failure to start top drive	FTS	1
X		Failure to stop on demand	Failure to stop top drive or incorrect shutdown process	STP	1
X	X	Internal leakage	Leakage internally of process or utility fluids	INL	3
	X	Leakage in closed position	Leakage through a valve (e.g. ram-valve) in closed position	LCP	2 (3)
X		High output	Output torque above specifications	HIO	2
X		Low output	Output torque below specifications	LOO	2
X		Noise	Excessive noise	NOI	3
X		Overheating	Overheating	OHE	3
X	X	Spurious operation	Unexpected operation	SPO	2
X		Structural deficiency	Material damages (cracks, wear, fracture, corrosion)	STD	3
X		Vibration	Excessive vibration	VIB	3 (2)
	X	Loss of redundancy	Loss of one or more redundancies (e.g. main control system, backup system)	LOR	2
	X	Loss of functions on both pods	Both pods are not functioning as desired	POD	1
	X	Plugged/choked	Choke or kill line plugged	PLU	3
	X	Fails to connect	Failure to connect upper connector	FCO	1
	X	Fails to disconnect	Failure to disconnect upper connector	FTD	1
X	X	Minor in-service problems	Loose items, discoloration, dirt	SER	3
X	X	Other	Failure modes not covered above	OTH	—
X	X	Unknown	Too little information to define a failure mode	UNK	—

^a See Table A.4. The codes shown apply to equipment classes marked with "X".

^b A proposed abbreviated code for the failure mode.

^c One of the three failure-mode types listed below; depending on type of failure, more than one of these categories can apply (e.g. a severe leakage can lead to stoppage of the equipment):

- 1) desired function is not obtained (e.g. failure to start);
- 2) specified function lost or outside accepted operational limits (e.g. spurious stop, high output);
- 3) failure indication is observed, but there is no immediate and critical impact on equipment-unit function. These are typically non-critical failures related to some degradation or incipient fault condition.